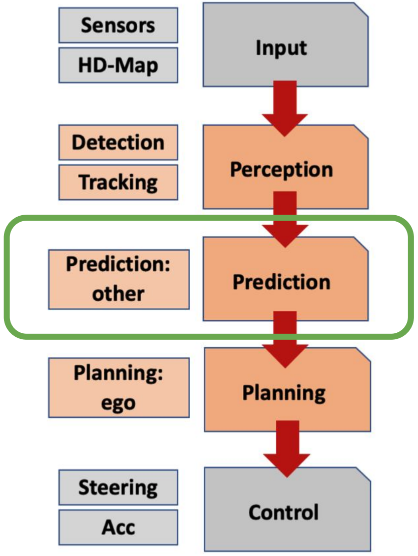
LAformer Explainability

Sri Nitchith Akula

TL;DR



Autonomous-Driving Stack

Motivation: In autonomous driving, the standard pipeline follows a P3 architecture (Perception, Prediction, Planning). The trajectory prediction module, which forecasts where other road users will go, is critical for safe planning, but these models are typically black boxes. Understanding **why** a model predicts a certain trajectory is essential for debugging failures and building trust.

Approach: Apply perturbation-based explainability (agent ablation, lane ablation, temporal occlusion) to LAformer, a lane-aware transformer.

Key Results:

- Lanes are the dominant feature (removing lanes shifts FDE more than compared to removing agents)
- Leave-one-out ablation has limits: found cases where no single agent matters, but removing all agents together causes a large shift
- Built an interactive Gradio demo for real-time "what-if" exploration

Code: https://github.com/nitchith/LAformer-Explainability

Problem Statement

Why explainability for trajectory prediction?

- Autonomous vehicles rely on trajectory prediction to anticipate other road users
- Modern models (GNNs, Transformers) achieve strong accuracy but are opaque
- For safety-critical deployment, we need to understand:

Question	Why it matters
Which agents influence the prediction?	Identify safety-critical interactions
How important are lanes vs. social context?	Validate architectural design choices
Which moments in history matter most?	Understand temporal dependencies

Literature Review: Trajectory Prediction

Model	Year	Venue	Key Idea
Social LSTM	2016	CVPR	Learned social pooling over neighbors
Social GAN	2018	CVPR	Adversarial multi-modal prediction
VectorNet	2020	CVPR	Polyline-based vectorized scene encoding
Trajectron++	2020	ECCV	Conditional VAE + interaction graph
QCNet	2023	CVPR	Query-centric encoding + two-stage decoder
Wayformer	2023	ICRA	Factored attention over agents and road
MTR	2023	NeurIPS	Motion Transformer with intention points
LAformer	**2024**	**CVPR**	**Lane-aware Transformer with lane scoring**
HPNet	2024	CVPR	Historical prediction attention for stability
GoalFlow	2025	CVPR	Goal-driven flow matching for multimodal trajectories

Trend: Increasing use of map/lane features alongside social context, with Transformer architectures dominating recent SOTA.

Literature Review: Explainability in ML

Common post-hoc methods:

- LIME (Ribeiro et al., 2016) => local surrogate models
- SHAP (Lundberg & Lee, 2017) => Shapley value attribution
- Gradient-based => Saliency maps, GradCAM

Why perturbation-based ablation for trajectory prediction?

- Input is a structured graph (polylines), not images => Might be hard to use gradient methods
- Helps with answering: What happens if this agent/lane wasn't there?
- Model-agnostic i.e works on any prediction model without architecture changes
- Computationally feasible: LAformer inference is ~0.2s per sample on CPU

LAformer Architecture

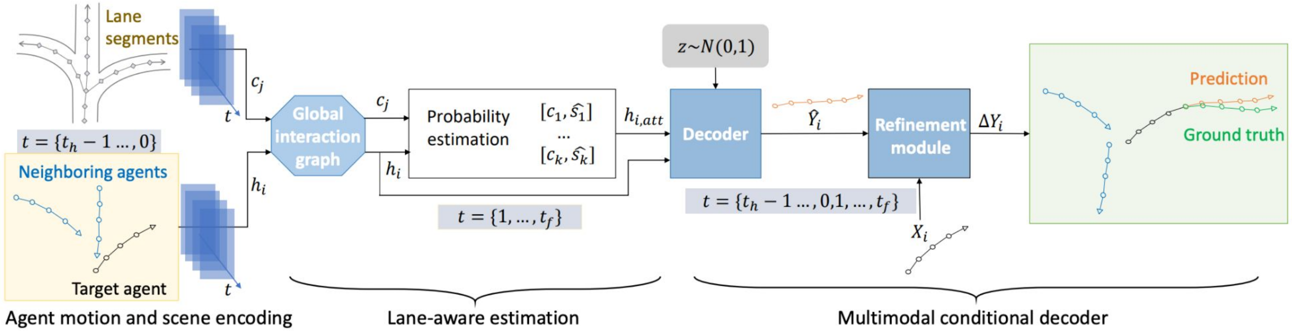


Figure 2. The LAformer framework takes vectorized trajectories and HD map lane segments as input. The agent motion and scene encodings are represented as h_i and c_j, respectively, and are later fused by the attention-based global interaction graph. The decoder then takes the target agent's past trajectory h_i, the updated motion information aligned with the candidate lane information $h_{i,\text{att}}$ from the lane-aware estimation module, and a random latent variable z as inputs, and predicts the future trajectory $\hat{Y}_i$. The refinement module further reduces the predicted offset ΔY to improve prediction accuracy.

Approach

1. Neighbor-Agent Ablation (Leave-One-Out)

Remove each agent's polyline individually, measure ADE/FDE shift from baseline.

2. Group Ablation

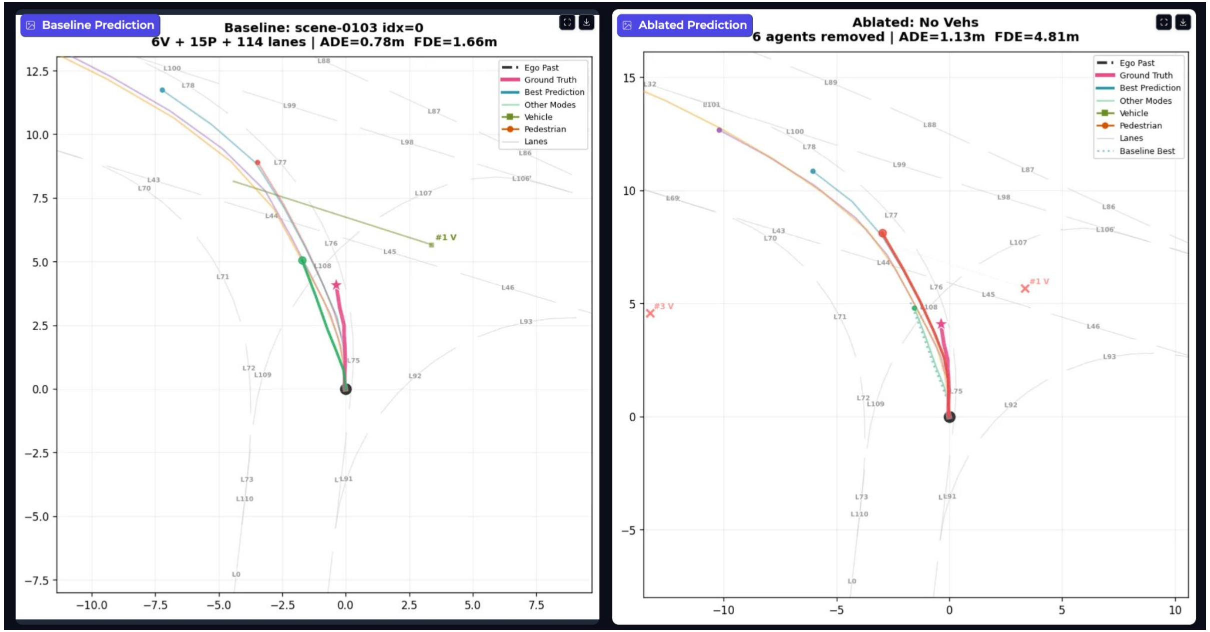
Remove entire categories and compare shifts:

- No/All Lanes
- No/All Vehicles
- No/All Pedestrians
- Only Ego vehicle

3. Temporal Occlusion

Mask each ego history timestep individually to find which moments matter most.

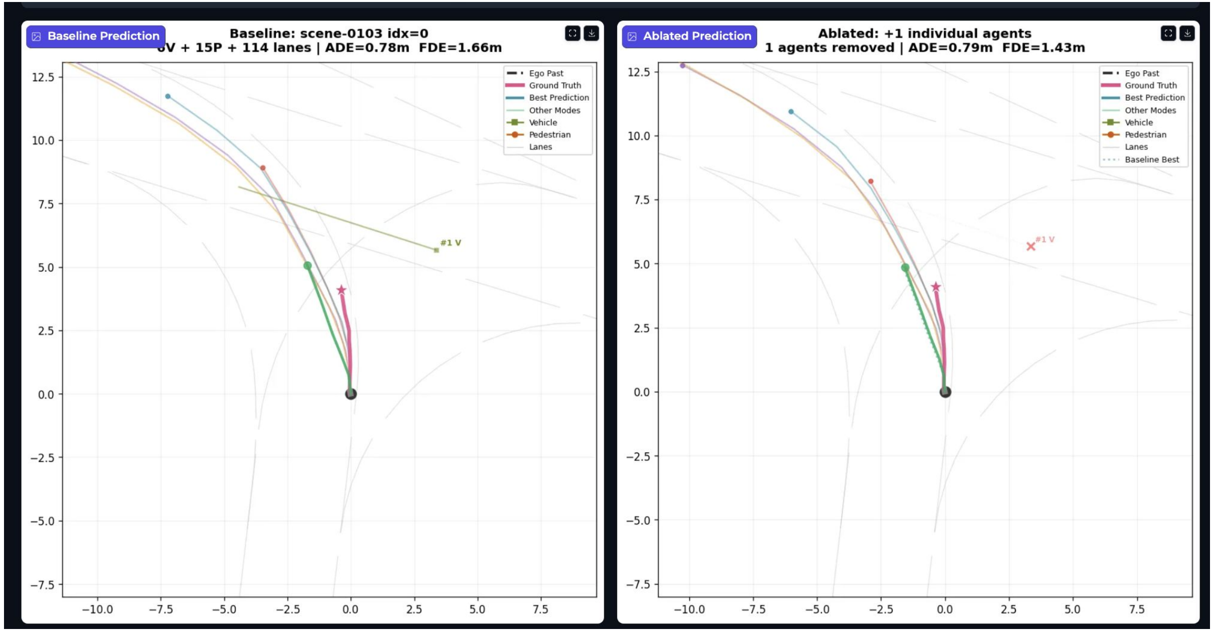
Results (All Vehicle Masking)



By removing all the other vehicles in the scene, we can see that the most-likely prediction is moving further.

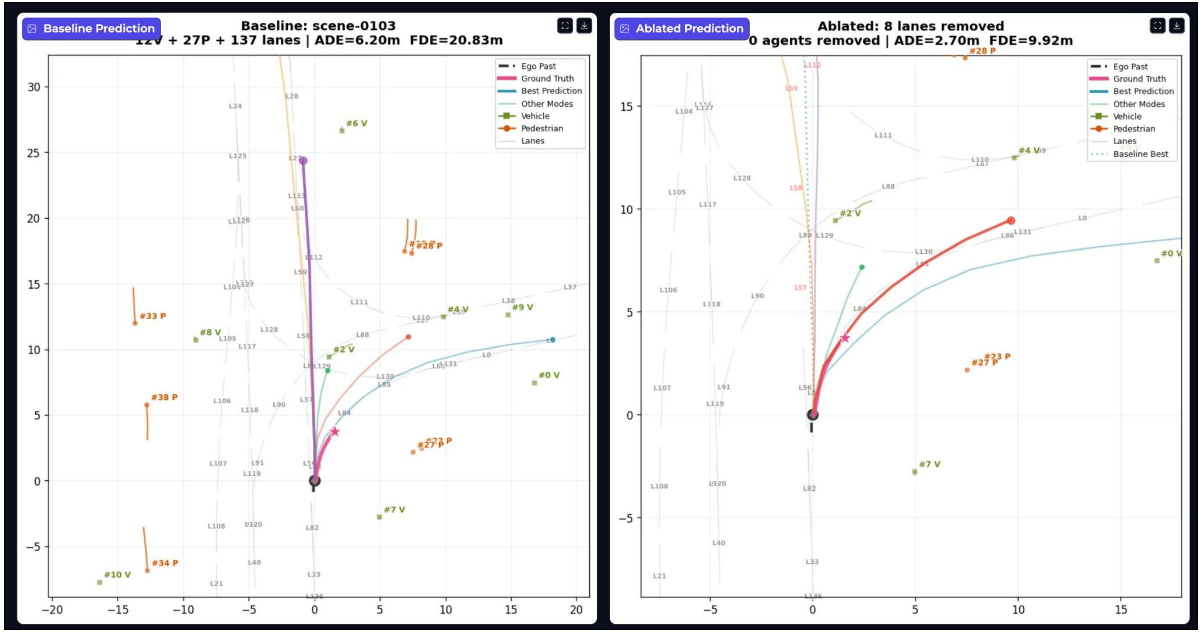
This can indicate that the model might be yielding for the vehicle #1 – but that’s not the case as seen in the next slide

Results (Masking Veh #1)



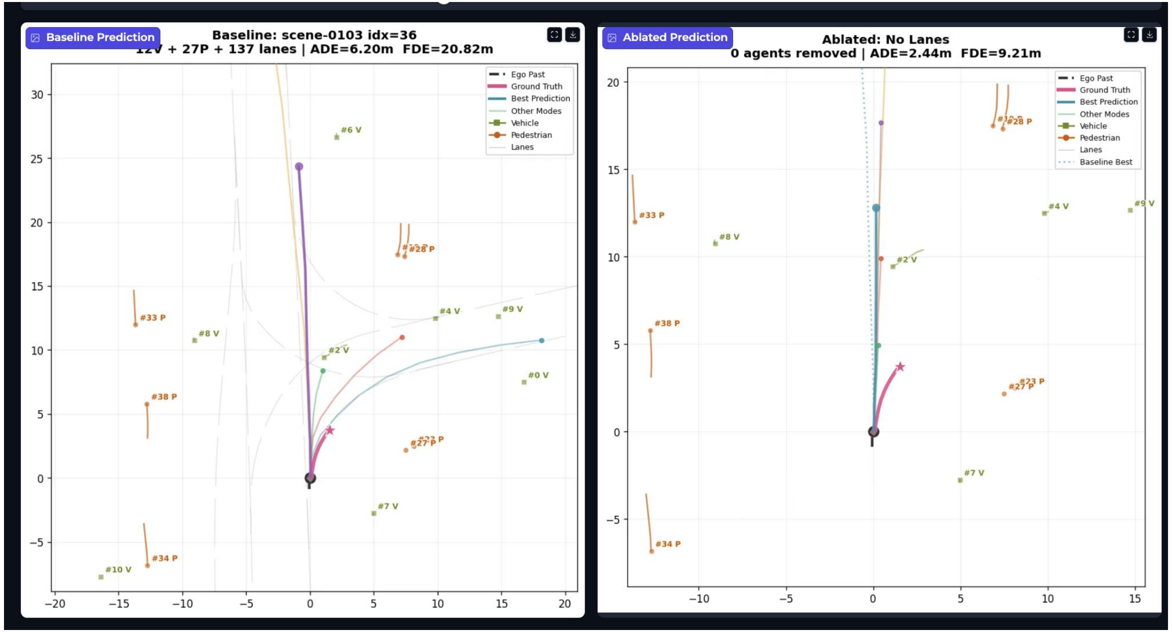
Just making for agent #1, we don’t see the best prediction to be proceeding

Results (Partial Lane Masking)



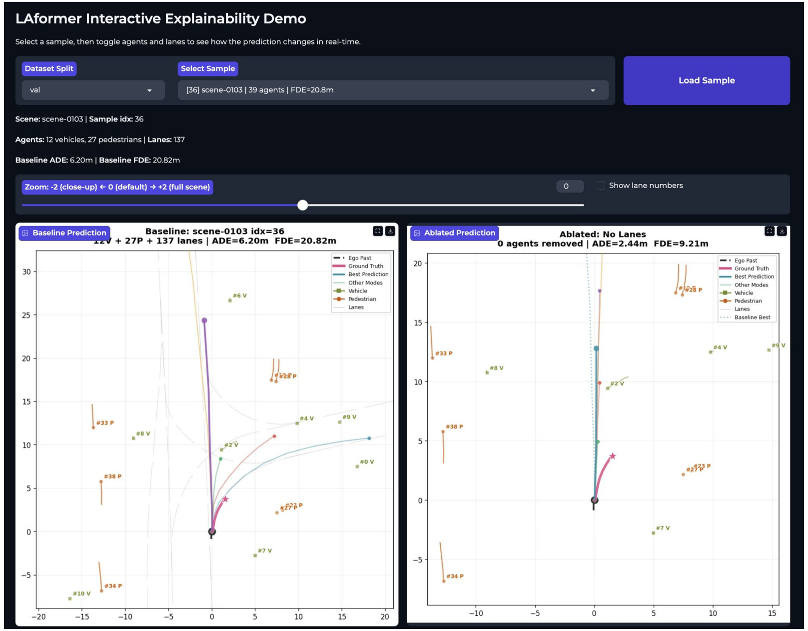
By removing the straight going lanes, the most-likely predictions is now turing

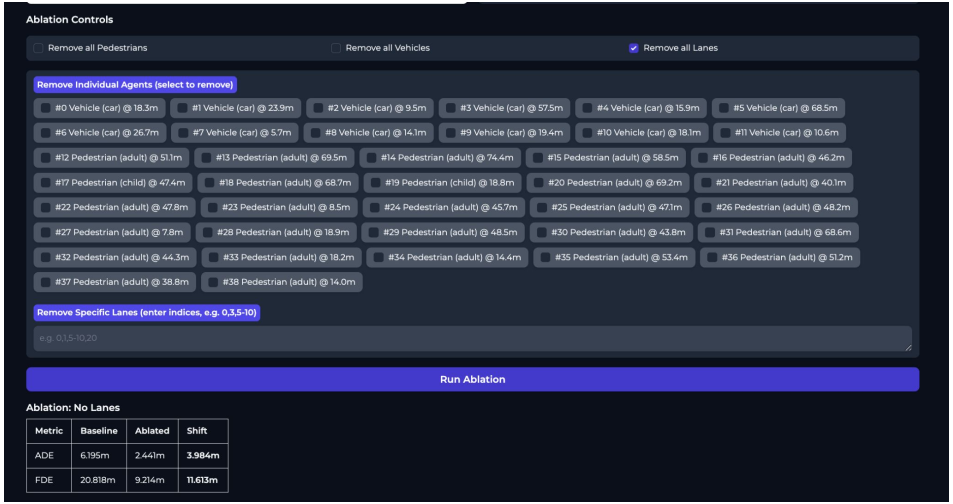
Results (Full Lane Masking)



By removing all the lanes, the model seems to rely more on its own history to go straight

Demo UI Screenshots





Metric	Baseline	Ablated	Shift
ADE	6.195m	2.441m	**3.984m**
FDE	20.818m	9.214m	**11.613m**

Demo

Youtube link: https://youtu.be/mDdcummOW_s

Code: https://github.com/nitchith/LAformer-Explainability

Conclusion

- Case study examples have shown interesting prediction changes due to the input changes
- There are many examples where the predictions doesn’t seem accurate visually even without any agent interaction.
- Explainability analysis is only as meaningful as the model's baseline accuracy i.e, if predictions aren't lane-consistent, ablation results become harder to interpret.
- Apply this framework to models with stronger agent interaction and lane conditioning (e.g., MTR, QCNet) for more interpretable ablation results
- We only analyzed the nuScenes-mini dataset for analysis. Looking at examples from a bigger dataset can give even more interesting examples

References

- 1. Liu et al., "LAformer: Trajectory Prediction with Lane-Aware Scene Constraints," CVPR 2024
- 2. LAformer source code: https://github.com/mengmengliu1998/LAformer
- 3. Gao et al., "VectorNet: Encoding HD Maps and Agent Dynamics," CVPR 2020
- 4. Salzmann et al., "Trajectron++: Dynamically-Feasible Trajectory Forecasting," ECCV 2020
- 5. Alahi et al., "Social LSTM: Human Trajectory Prediction," CVPR 2016
- 6. Gupta et al., "Social GAN: Socially Acceptable Trajectories," CVPR 2018
- 7. Zhou et al., "QCNet: Query-Centric Trajectory Prediction," CVPR 2023
- 8. Tang et al., "HPNet: Dynamic Trajectory Forecasting with Historical Prediction Attention," CVPR 2024
- 9. Xing et al., "GoalFlow: Goal-Driven Flow Matching for Multimodal Trajectories," CVPR 2025
- 10. Caesar et al., "nuScenes: A Multimodal Dataset for Autonomous Driving," CVPR 2020
- 11. Ribeiro et al., "LIME: Why Should I Trust You?," KDD 2016
- 12. Lundberg & Lee, "SHAP: A Unified Approach to Interpreting Model Predictions," NeurIPS 2017